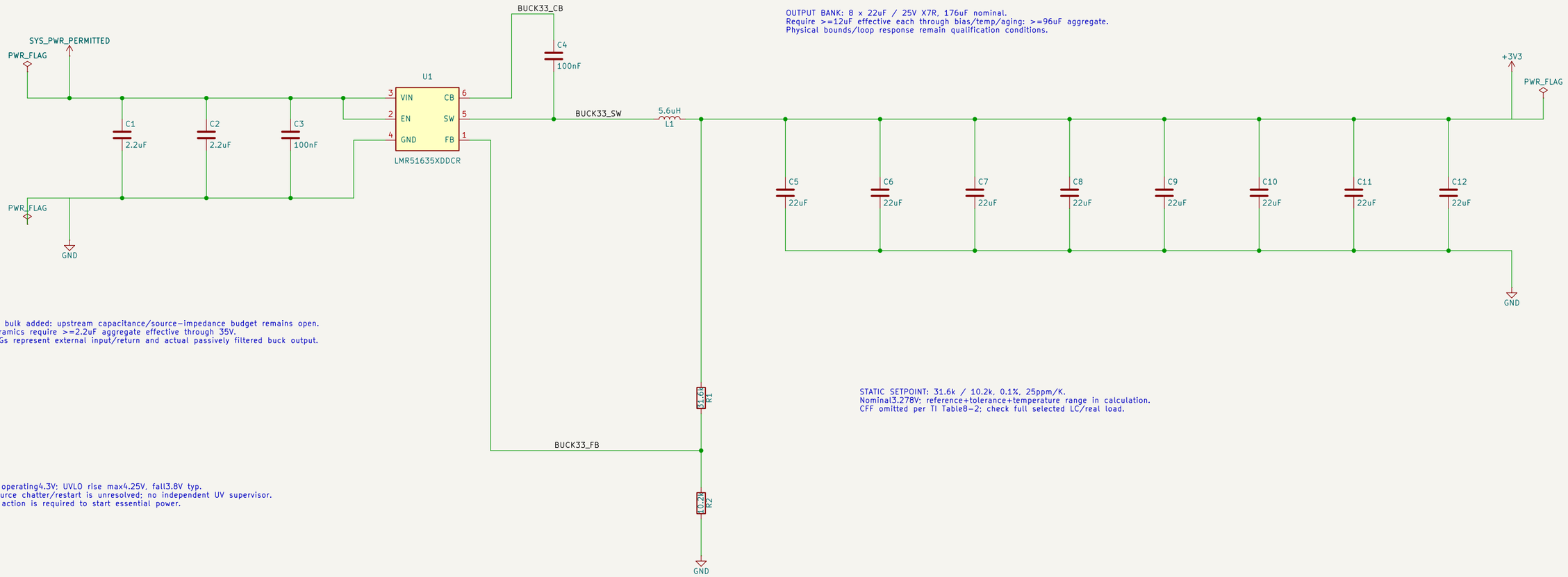
ESSENTIAL 3V3 — LMR51635XDDCR DEVELOPMENT

Input after RBF/deployment and source OV permission; autonomous EN = VIN.
5..35 V development envelope; 4.5 V sensitivity. 2 A is a sizing target, not qualified load.



No input bulk added: upstream capacitance/source-impedance budget remains open.
Input ceramics require >=2.2uF aggregate effective through 35V.
PWR_FLAGs represent external input/return and actual passively filtered buck output.

STATIC SETPOINT: 31.6k / 10.2k, 0.1%, 25ppm/K.
Nominal3.278V; reference+tolerance+temperature range in calculation.
CFF omitted per TI Table8-2; check full selected LC/real load.

VIN min operating4.3V; UVLO rise max4.25V, fall3.8V typ.
Weak-source chatter/restart is unresolved; no independent UV supervisor.
No MCU action is required to start essential power.

OPEN FAULT CONTAINMENT: a failed-short high-side switch can expose +3V3
consumers and 25V output bank to ~35V. Autonomous 3V3 OV isolation is not present.
Upstream solar OV protection does not protect the 3V3 rail from this failure.

POLICY OPEN: S03 native severity persistence and S11 visible rating-field styling unavailable.
Existing custom U1 reused; original S08 review-marker coverage unresolved. No PCB work.

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